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Practice Exam 1

Discrete Structures

28th day of September of the year of our Lord 2026

1 *Answer the questions below by selecting exactly one answer for each.*

1. Let φ be a *wff*. Which of the following are equivalent to $\forall n(\varphi(n))$?[1]

[1] This question has *two* correct answers.

- ● $\varphi(0) \wedge \forall k(\varphi(k) \Rightarrow \varphi(k+1))$
- ○ $\forall k(\forall j(j < k \Rightarrow \varphi(j)) \Rightarrow \varphi(k+1))$
- ● $\varphi(0) \wedge \neg\exists k(\varphi(k) \wedge \neg\varphi(k+1))$
- ○ $\forall k(\forall j(j < k \Rightarrow \varphi(j)) \Rightarrow \varphi(k))$

$\equiv \forall K\,(\neg(\varphi(K) \wedge \neg\varphi(K+1)))$
$\equiv \forall K\,(\neg\varphi(K) \vee \varphi(K+1))$
$\equiv \forall K\,(\varphi(K) \rightarrow \varphi(K+1))$

2. Which of the following propositions ***contradicts*** the axioms of arithmetic?

- ○ $\exists x \forall y(\mathrm{suc}(y) \neq x)$
- ● $\forall x \exists y(\mathrm{suc}(y) = x)$
- ○ $\forall x \exists y \exists z(x + z = y)$
- ○ $\exists x \forall y \exists z(x + z = y)$

$\forall x \exists y\,(\mathrm{suc}(y) = x)$ means every number is a successor; but 0 is not a successor.

3. If p and q are propositions, which of the following can be inferred from $p \vee q$?

- ○ $p \rightarrow \neg q$
- ○ $\neg p \wedge \neg q$
- ○ p
- ● $\neg q \rightarrow p$

$p \vee q \equiv p \vee (\neg\neg q) \equiv \neg(\neg q) \vee p \equiv \neg q \rightarrow p$.

4. If p and q are propositions, which of the following is equivalent to $\neg(p \rightarrow q)$?

- ● $p \wedge \neg q$
- ○ $p \vee \neg q$
- ○ $\neg p \rightarrow \neg q$
- ○ $\neg q \rightarrow \neg p$

$\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \equiv (\neg\neg p) \wedge \neg q \equiv p \wedge q$.

5. If φ is a *wff* such that $\forall k(\varphi(k) \Rightarrow \varphi(k+2))$, which numbers must φ be *true* about?

- ○ *The even natural numbers*
- ○ *None of the natural numbers*
- ○ *All of the natural numbers*
- ● *Not enough information*

$\forall K(\varphi(K) \rightarrow \varphi(K+2))$ is not enough to prove $\varphi(n)$ for any n; however, this doesn't tell us that $\neg\varphi(n)$ holds for any particular n either. If we knew $\varphi(0)$, then we would be able to prove $\varphi(n)$ for every even n; if we knew $\neg\varphi(10)$, then we would be able to prove $\neg\varphi(n)$ for every even n less than 10.

6. This sentence is *false*.

- ○ *True*
- ○ *False*
- ● *Neither True nor False*
- ○ *Not enough information*

7. If this sentence is *false*, then $\exists m(3 = 2^m)$.

- ○ *True*
- ○ *False*
- ○ *Neither True nor False*
- ● *Not enough information*

This could be either T or ⊥ without causing a contradiction.

8. If 0 is a multiple of 2, then this sentence is *false*.

- ○ *True*
- ○ *False*
- ● *Neither True nor False*
- ○ *Not enough information*

9. In the theory of arithmetic, the symbol $\mathrm{suc}(\bigcirc)$ denotes a predicate.

- ○ *True*
- ● *False*
- ○ *Neither True nor False*
- ○ *Not enough information*

The suc symbol is a (unary) functional symbol; it maps objects to objects. Predicates map objects to truth values, not objects.

10. Strong induction is capable of proving more things than weak induction.

- ○ *True*
- ● *False*
- ○ *Neither True nor False*
- ○ *Not enough information*

$(\varphi(0) \wedge \forall K(\varphi(K) \rightarrow \varphi(K+1))) \Leftrightarrow (\forall K(\forall j(j<K \rightarrow \varphi(j)) \rightarrow \varphi(K)))$

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2 *Follow the instructions for each question below.*

$1 = n/2^k \Leftrightarrow k = \log_2(n)$

1. Find a closed form for the recurrence relation given below. Your closed form should not involve iterated operations. You do not need to justify your answer.

$$T(1) := 4$$
$$T(2 \cdot n) := 2 \cdot T(n) + n$$

When n is a power of 2, $T(n) = 4n + n \cdot \log_2(n)$.
In other words:

$$\forall n \left(\exists m (n = 2^m) \Rightarrow T(n) = 4 \cdot n + n \cdot \log_2(n) \right).$$

$T(n) = 2 \cdot T(n/2) + n$
$= 2(2 \cdot T(n/2^2) + n/2) + n$
$= 2^2 T(n/2^2) + 2n$
$= 2^2(2 \cdot T(n/2^3) + n/2^2) + 2n$
$= 2^3 T(n/2^3) + 3n$
$\vdots$
$= 2^k T(n/2^k) + k \cdot n$
$\vdots$
$= 2^{\log_2(n)} \cdot T(1) + \log_2(n) \cdot n$
$= n \cdot 4 + n \cdot \log_2(n)$

2. Let φ be a *wff* and consider the two propositions given below.

$$\forall k \left(\forall j (j < k \Rightarrow \varphi(j)) \Rightarrow \varphi(k) \right)$$
$$\forall k \left(\forall j (j \leqslant k \Rightarrow \varphi(j)) \Rightarrow \varphi(k+1) \right)$$

Explain why the first line is equivalent to $\forall n(\varphi(n))$ but the second line is not.

It should be clear that $\forall n(\varphi(n))$ implies both of the lines, since we could easilly prove $\varphi(k)$ (for the first line) and $\varphi(k+1)$ (for the second line) by universal elimination. So, let's look at the converse direction.
If we assume $\forall K (\forall j (j < K \Rightarrow \varphi(j)) \Rightarrow \varphi(K))$, then we know in particular, "when $K=0$," that $\forall j (j < 0 \Rightarrow \varphi(j)) \Rightarrow \varphi(0)$. Since $\forall x (x \geq 0)$, the promise of this inner conditional is false, so $\forall j (j < 0 \Rightarrow \varphi(j))$ is true! Applying Modus Ponens, this proves $\varphi(0)$. Now, "when $K=1$", our assumption says $\forall j (j < 1 \Rightarrow \varphi(j)) \Rightarrow \varphi(1)$; the premise $\forall j (j < 1 \Rightarrow \varphi(j))$ is equivalent to $0 < 1 \Rightarrow \varphi(0)$, which is equivalent to $\varphi(0)$, so our assumption actually says $\varphi(0) \Rightarrow \varphi(1)$. However, we just verified $\varphi(0)$! Therefore, $\varphi(1)$ by Modus Ponens. Continuing on in this way, we can prove $\varphi(n)$ for any natural number n; we could easily formalize this argument using Axiom 4: the schema of weak induction.

} we can perform these simplifications because 0 is the only natural number smaller than 1.

The second line says $\forall K (\forall j (j \leq K \Rightarrow \varphi(j)) \Rightarrow \varphi(K+1))$; assuming this and setting "$K=0$", we get $\forall j (j \leq 0 \Rightarrow \varphi(j)) \Rightarrow \varphi(1)$, which simplifies to $\varphi(0) \Rightarrow \varphi(1)$. When we simplify the expression at "$K=1$", it says $(\varphi(0) \wedge \varphi(1)) \Rightarrow \varphi(2)$. We get all the same implicative statements, but we never get $\varphi(0)$! So we can't apply Modus Ponens, and we fail to get $\varphi(n)$ for any n.

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3 *For this problem, you may rely on any result of arithmetic we have studied. You may carry out calculations involving addition, multiplication, and exponentiation of specific numbers without justification.*

We define a natural number x to be *"even"* when $\exists y(x = 2 \cdot y)$. We similarly define a natural number x to be *"odd"* when $\exists y(x = 2 \cdot y + 1)$.

Prove that $n \cdot (n + 1)$ is even for every natural number n.

"Gauss' Little Theorem"

Proof 1: Let n be an arbitrary natural number and recall the theorem that says $2 \cdot \sum_{i=0}^{n} i = n \cdot (n+1)$. Since n is a natural number, we know $\sum_{i=0}^{n} i$ is also a natural number, and thus $\exists m(2 \cdot m = n \cdot (n+1))$. Therefore, by definition, $n \cdot (n+1)$ is even. QED

Proof 2: We will show $\forall n(\exists m(2m = n(n+1)))$ by (weak) induction.

Basis Step: We will show $\exists m(2m = 0 \cdot (0+1))$. Observe $0 \cdot (0+1) = 0 \cdot 1 = 0 = 2 \cdot 0$, so $\exists m(2 \cdot m = 0 \cdot (0+1))$.

Inductive Step: Let K be a natural number. Assume $\exists m(2m = K \cdot (K+1))$. We will show that $\exists m(2m = (K+1)((K+1)+1))$.

By the inductive hypothesis, we know there exists some natural number x such that $2x = K \cdot (K+1)$. Now, observe the following.

$$\begin{aligned}
(K+1)((K+1)+1) &= (K+1)(K+(1+1)) && \text{by associativity of addition} \\
&= (K+1)(K+2) \\
&= (K+1) \cdot K + (K+1) \cdot 2 && \text{by distributivity of multiplication over addition} \\
&= K \cdot (K+1) + 2 \cdot (K+1) && \text{by commutativity of multiplication} \\
&= 2 \cdot x + 2 \cdot (K+1) && \text{by the inductive hypothesis} \\
&= 2 \cdot (x + (K+1)) && \text{by distributivity of multiplication over addition}
\end{aligned}$$

Since x and K are both natural numbers, $x + (K+1)$ is also a natural number. Therefore, $\exists m(2m = (K+1)((K+1)+1))$.

Therefore, by Axiom 4, we conclude that $\forall n(\exists m(2m = n \cdot (n+1)))$, which means precisely that $n \cdot (n+1)$ is even for every natural number n.

QED

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4 *For this problem, you may rely on any result of arithmetic we have studied. You may carry out calculations involving addition, multiplication, and exponentiation of specific numbers without justification.*

Given two arbitrary natural numbers x and y, we say that "*x divides y*" when we can express y as a multiple of x—we define this formally below.

$$x \mid y \;:\Leftrightarrow\; \exists m(x \cdot m = y)$$

Use any valid form of induction to show that $4 \mid 9^n + 3$ for every natural number n.

Proof: We will prove $\forall n(4 \mid 9^n+3)$ by strong induction (just so you see what that looks like).

Let K be an arbitrary natural number.

Assume $\forall j(j<K \Rightarrow 4 \mid 9^j+3)$. We now prove $4 \mid 9^K+3$ by taking two cases.

Case 1: Suppose $K=0$. We can see $9^0+3 = 1+3 = 4 = 4\cdot 1$, so $\exists m(4\cdot m = 9^0+3)$.
Thus, $4 \mid 9^0+3$ by definition.

Case 2: Suppose $K \neq 0$, which implies $K = j+1$ for some natural number j.
Since $K=j+1$, we know $K>j$ by definition, and thus $4 \mid 9^j+3$.
This implies there exists some m such that $4m = 9^j+3$ by definition.
Now, we observe the following deduction:

$9^K+3 = 9^{j+1}+3$ by [redacted] our earlier result: $K = j+1$
$= 9^j \cdot 9 + 3$ by definition of exponentiation
$= ((9^j \cdot 9 + 3) + 24) - 24$ by the theorem $\forall x \forall y((x+y)-y = x)$.
$= (9^j \cdot 9 + (3+24)) - 24$ by associativity of addition
$= (9^j \cdot 9 + 3 \cdot 9) - 4 \cdot 6$
$= (9 \cdot 9^j + 9 \cdot 3) - 4 \cdot 6$ by commutativity of multiplication
$= 9 \cdot (9^j + 3) - 4 \cdot 6$ by distributivity of multiplication over addition
$= 9 \cdot (4 \cdot m) - 4 \cdot 6$ by the inductive hypothesis
$= (4 \cdot m) \cdot 9 - 4 \cdot 6$ by commutativity of multiplication
$= 4 \cdot (m \cdot 9) - 4 \cdot 6$ by associativity of multiplication
$= 4 \cdot (m \cdot 9 - 6)$ by distributivity of multiplication over subtraction

Since m is a natural number, so is $m \cdot 9 - 6$, so $\exists l(4 \cdot l = 9^K+3)$.
This shows $4 \mid 9^K+3$ [redacted] as desired.

Therefore, by the theorem of strong induction, $\forall n(4 \mid 9^n+3)$.

QED

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5 *For this problem, you may only assume the statements written on the side.*

Prove that multiplication distributes over addition by demonstrating the following.

$$\forall x \forall y \forall z (x \cdot (y + z) = (x \cdot y) + (x \cdot z))$$

Proof: Let x and y be natural numbers. We will prove $\forall z\,(x\cdot(y+z)=(x\cdot y)+(x\cdot z))$ by (weak) induction.

Basis Step: We will show $x\cdot(y+0) = (x\cdot y) + (x\cdot 0)$.

Observe $x\cdot(y+0) = x\cdot y$ by definition of addition

$= (x\cdot y)+0$ by definition of addition

$= (x\cdot y)+(x\cdot 0)$ by definition of multiplication

Therefore, $x\cdot(y+0) = (x\cdot y)+(x\cdot 0)$.

Inductive Step: Let k be a natural number. Assume $x(y+k) = (x\cdot y) + (x\cdot k)$.

We will now show $x\cdot(y+\text{suc}(k)) = (x\cdot y) + (x\cdot \text{suc}(k))$.

Observe: $x\cdot(y+\text{suc}(k)) = x\cdot \text{suc}(y+k)$ by definition of addition

$= x(y+k) + x$ by definition of multiplication

$= \big((x\cdot y)+(x\cdot k)\big)+ x$ by the inductive hypothesis

$= (x\cdot y)+\big((x\cdot k)+x\big)$ by associativity of addition

$= (x\cdot y) + (x\cdot \text{suc}(k))$ by definition of multiplication

Therefore, $x\cdot(y+\text{suc}(k)) = (x\cdot y)+(x\cdot \text{suc}(k))$ as required.

Therefore, we conclude $\forall\tilde{x}\forall\tilde{y}\forall\tilde{z}\big(\tilde{x}\cdot(\tilde{y}+\tilde{z}) = (\tilde{x}\cdot\tilde{y})+(\tilde{x}\cdot\tilde{z})\big)$ by Axiom 4.

QED

In this problem, you may assume all of the axioms of the theory of arithmetic.

0. $\exists x(x = 0)$.
1. $\forall x(\text{suc}(x) \neq 0)$
2. $\forall x \exists y(y = \text{suc}(x))$
3. $\forall x \forall y(\text{suc}(x) = \text{suc}(y) \Rightarrow x = y)$
4. The Schema of Weak Induction

Additionally, you may assume the following theorems about addition.

5. $\forall x(x + 1 = \text{suc}(x))$
7. $\forall x \forall y \forall z(x + (y + z) = (x + y) + z)$
8. $\forall x \forall y(x + y = y + x)$

You may also assume the following theorems about multiplication.

9. $\forall x(0 \cdot x = 0)$
10. $\forall x(x \cdot 1 = x) \wedge \forall x(1 \cdot x = x)$